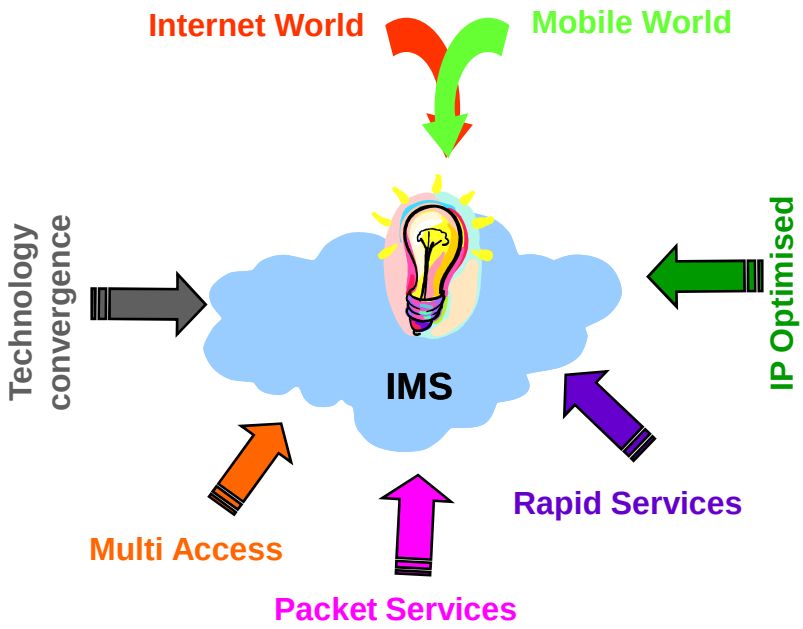
IP Multimedia Service example:

IP Multimedia Subsystem

What is IMS?

- IP Multimedia Subsystem (IMS) is an architecture for offering services on the packet domain.
- Not just Voice over IP - IMS is a generic architecture for offering **multimedia services**.
- IMS is *access agnostic*, hence it is not just for UMTS or GPRS, but will also support WLAN, fixed line, etc.



What is IMS (IP-Multimedia Subsystem)? (1)

- The **IP Multimedia Subsystem** (**IMS**) is an architectural framework for delivering internet protocol (IP) multimedia to mobile users.

- It was originally designed by the wireless standards body 3rd Generation Partnership Project (3GPP), and is part of the vision for evolving mobile networks beyond GSM.

- Its original formulation (3GPP R5) represented an approach to delivering "Internet services" over GPRS. This vision was later updated by 3GPP, 3GPP2 and TISPAN by requiring support of networks other than GPRS, such as Wireless LAN, CDMA2000 and fixed line.

What is IMS (IP-Multimedia Subsystem)? (2)

- A new IP based architecture that enables convergence of -
  - Data
  - Voice
  - Mobile Network Technology

- Fill the gap between -
  - Traditional telecommunications technology and
  - Internet technology

- IMS offers both -
  - Service convergence
  - Network convergence

Relation with other services

- IMS is not related to other services (e.g., streaming, messaging, presence, etc.)

  - Because IMS is not a service
  - But IMS is a **service enabler**
  - So all the services (e.g., streaming, multimedia messaging, presence) can be delivered by using IMS

What does IMS give? 1/2

- **Person-to-person communications** aided by the always on model.
- **Person-to-machine communications**
- **Machine-to-machine communications (e.g., telemetry)**
- **Convergence** of all media communications on to the packet network.
- **Multimedia communications** based on a single solution.

IMS: The truth…

All IMS services can be done without IMS

- This is true! – Other technologies can be used. The end-user probably doesn't care about the enabling technologies
- However – Without IMS, service creation can be complex, and expensive to the operator and the user
- The key is integrating different services (e.g., multimedia, presence, instant messaging, web browsing, location, personalised services) within a single technology

IMS: The truth…

IMS creates operator-bypass

- The operator can be used as a bit-pipe for accessing services from elsewhere
- IMS does not introduce this issue – it is perfectly feasible today for an end-user to use GPRS to obtain an IP address, access the Internet and access cheaper, richer services
- IMS needs to offer quality

What does IMS give? 2/2

- Significant reduction in the **cost of networks** both in terms of capex and opex by supporting both native IP based services and voice services on IP
- **Service control** by introducing service signalling
- **Access agnostic**
- Integrated **charging architecture**
- **Personalised services** are shorter lived or need to be tailored for smaller groups of individuals. Rapid service creation is vital.

Layered View of IMS (1)

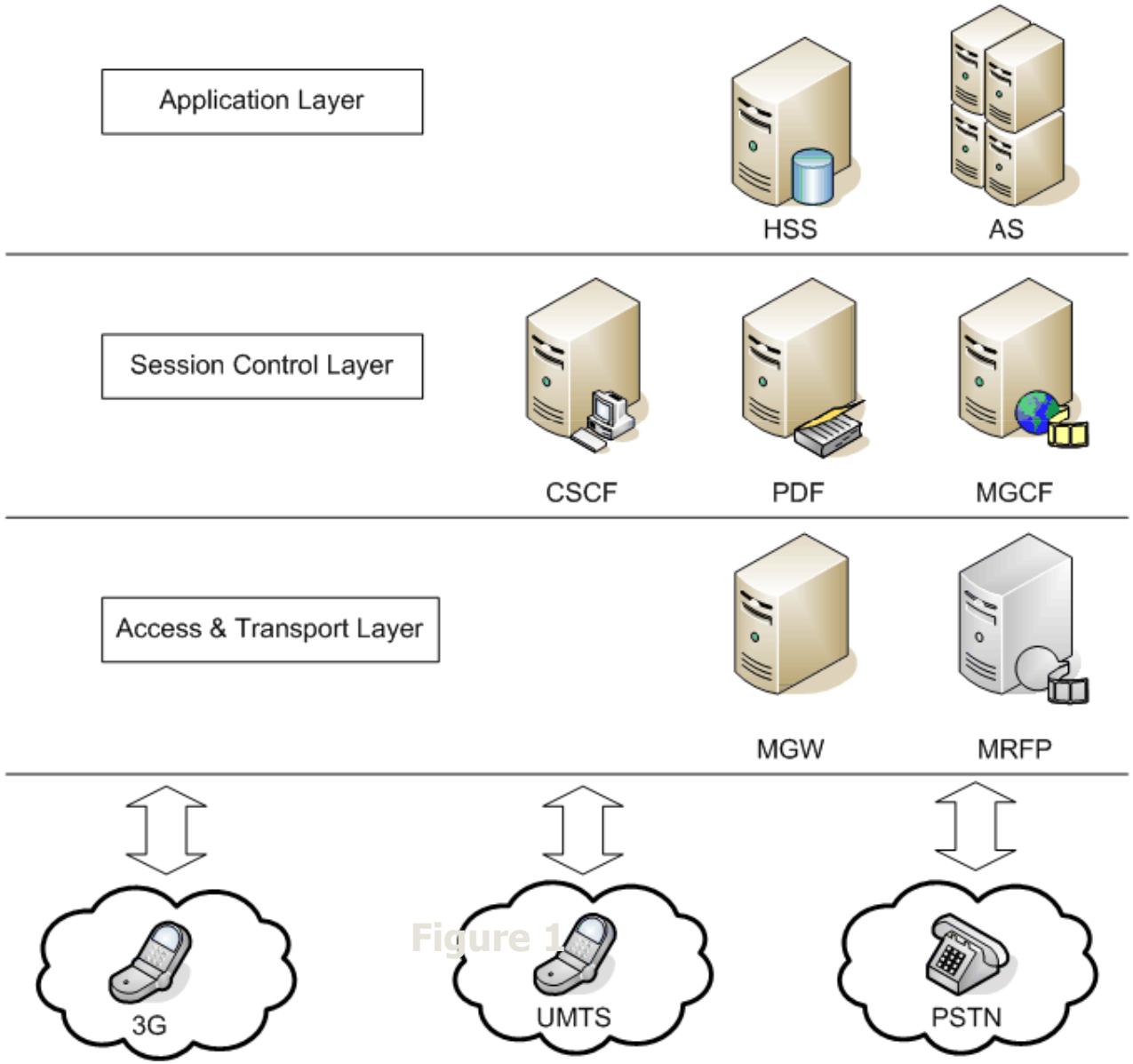
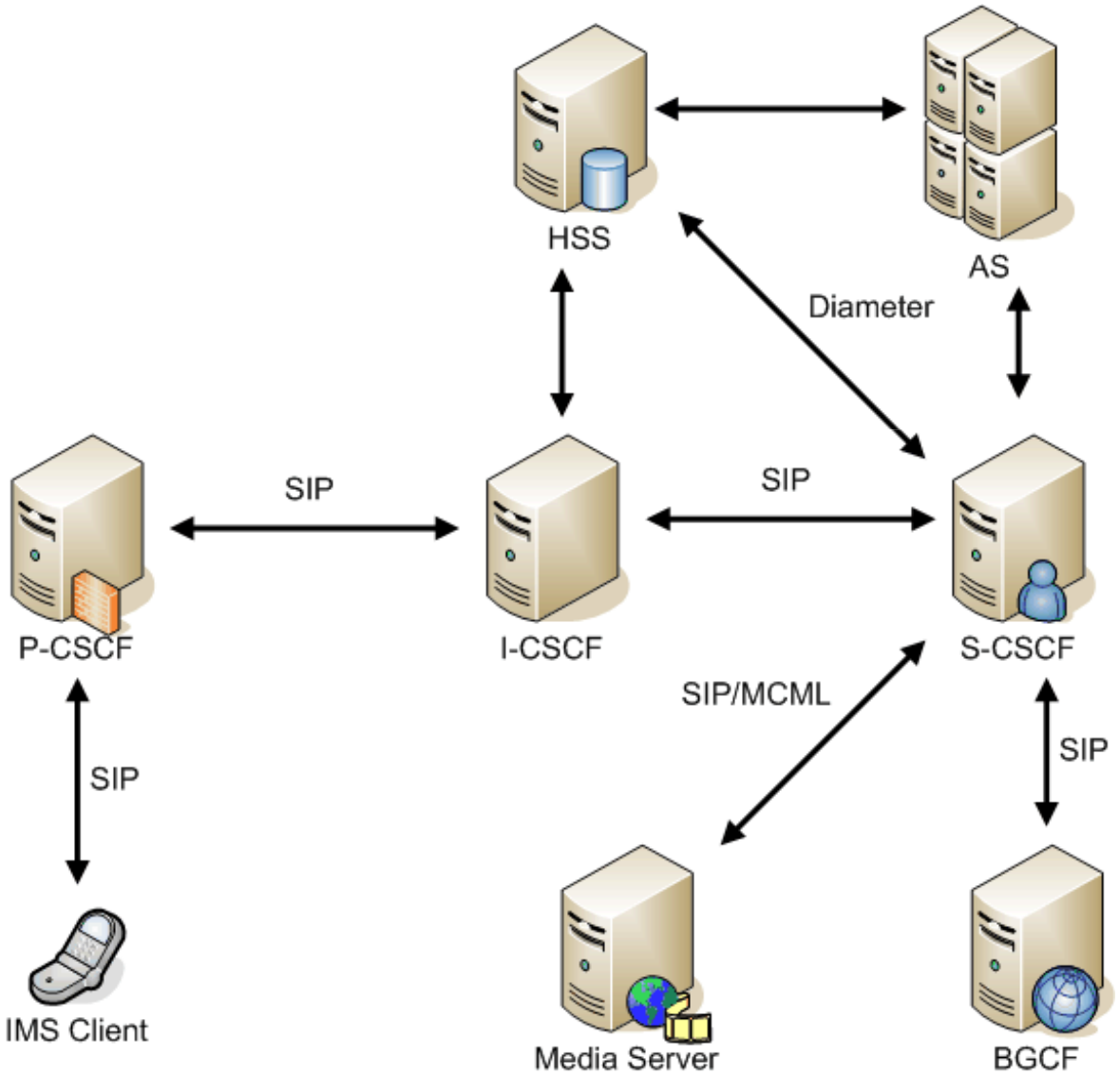


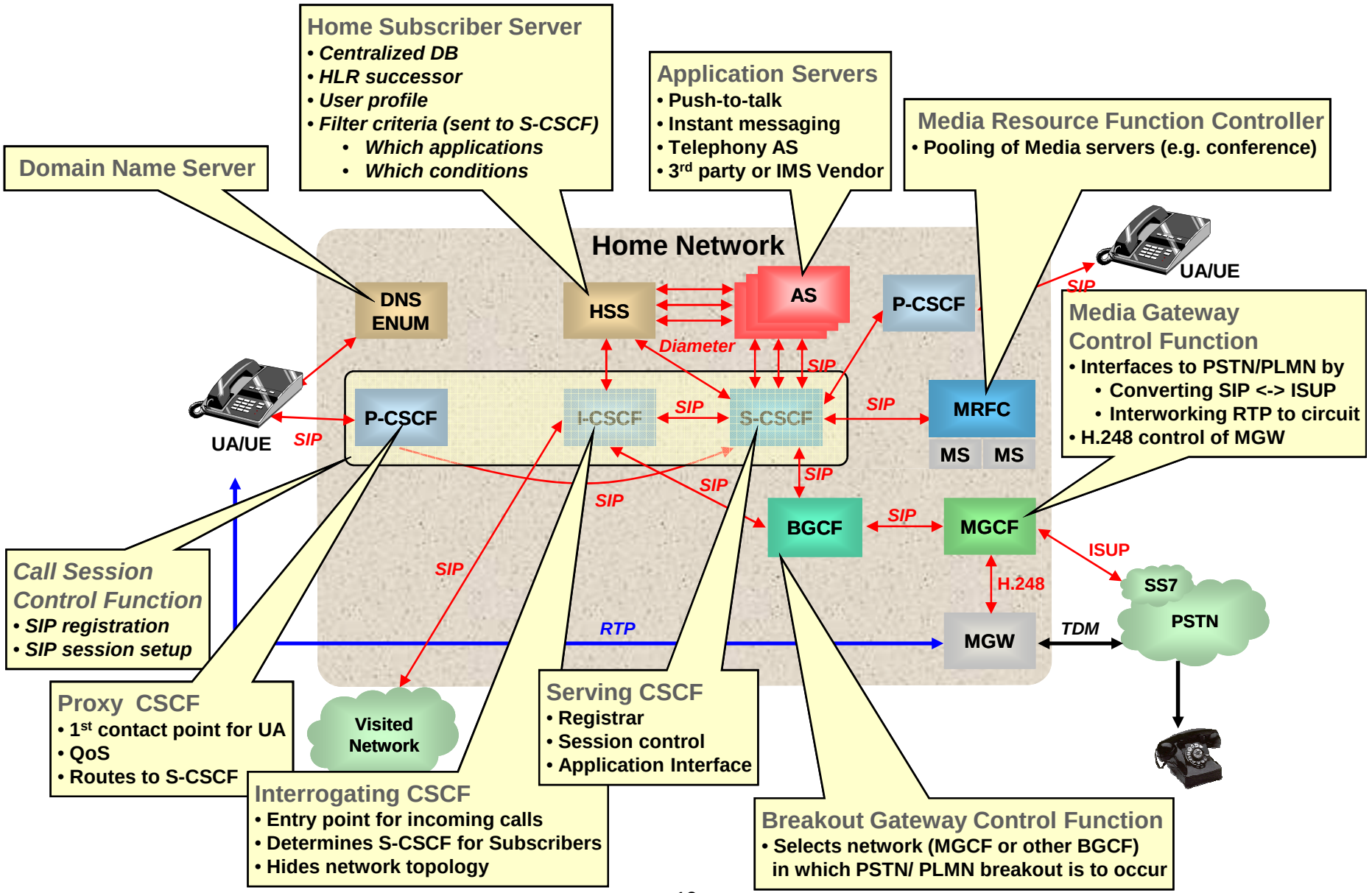
Figure 1

Functional Components of IMS (1)

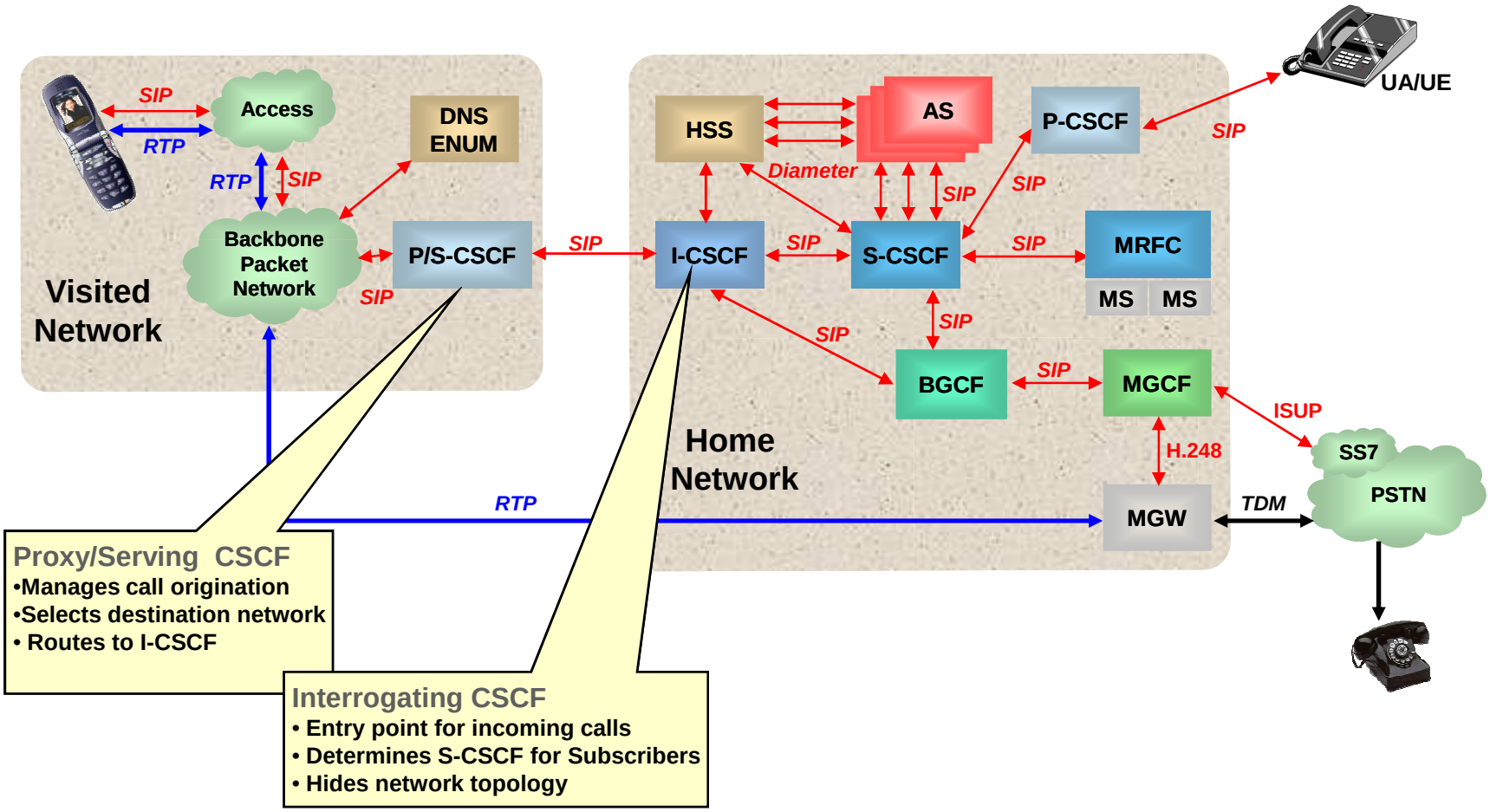


- P-CSCF (Proxy-CSCF)
  - Entry point to IMS for devices (Both Home & Visited)
  - Forward SIP messages to the S-CSCF
- I-CSCF (Interrogating-CSCF)
  - Entry point to IMS from other networks
- S-CSCF (Serving-CSCF)
  - Provides Session control services for the IMS client
  - Maintains a session state as needed by the network operator
- HSS (Home Subscriber Service)
  - Information about the end users and the services
- BGCF (Breakout gateway control function)
  - Selects the network in case of PSTN breakout
  - Selects a MGCF, for inter-working with the PSTN

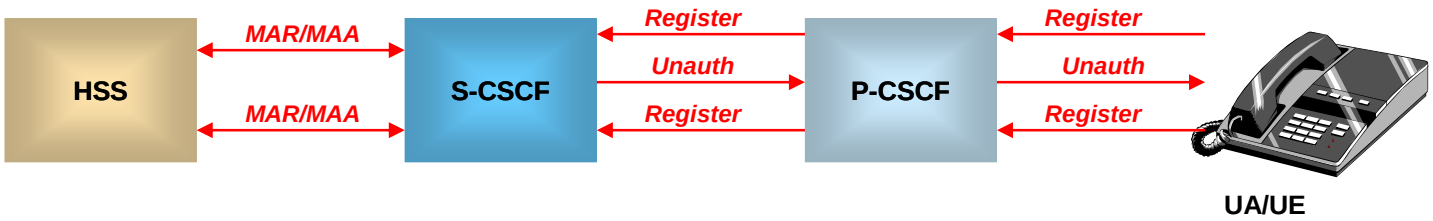
IMS Home Network - Functional Elements



IMS Network-to-Network Connectivity

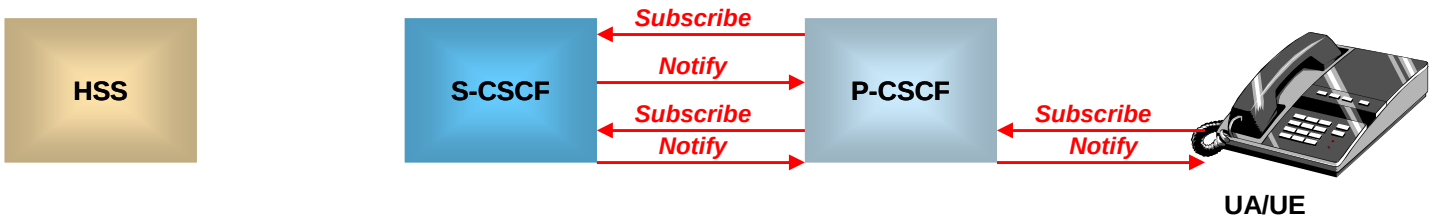


IMS UE Registration



- The UA/UE Registers with the S-CSCF
- The S-CSCF consults HSS for Authentication
- The S-CSCF Challenges the UA/UE
- The UA/UE Registers with Credentials to the S-CSCF
- S-CSCF Authenticates with HSS and Downloads User Profile

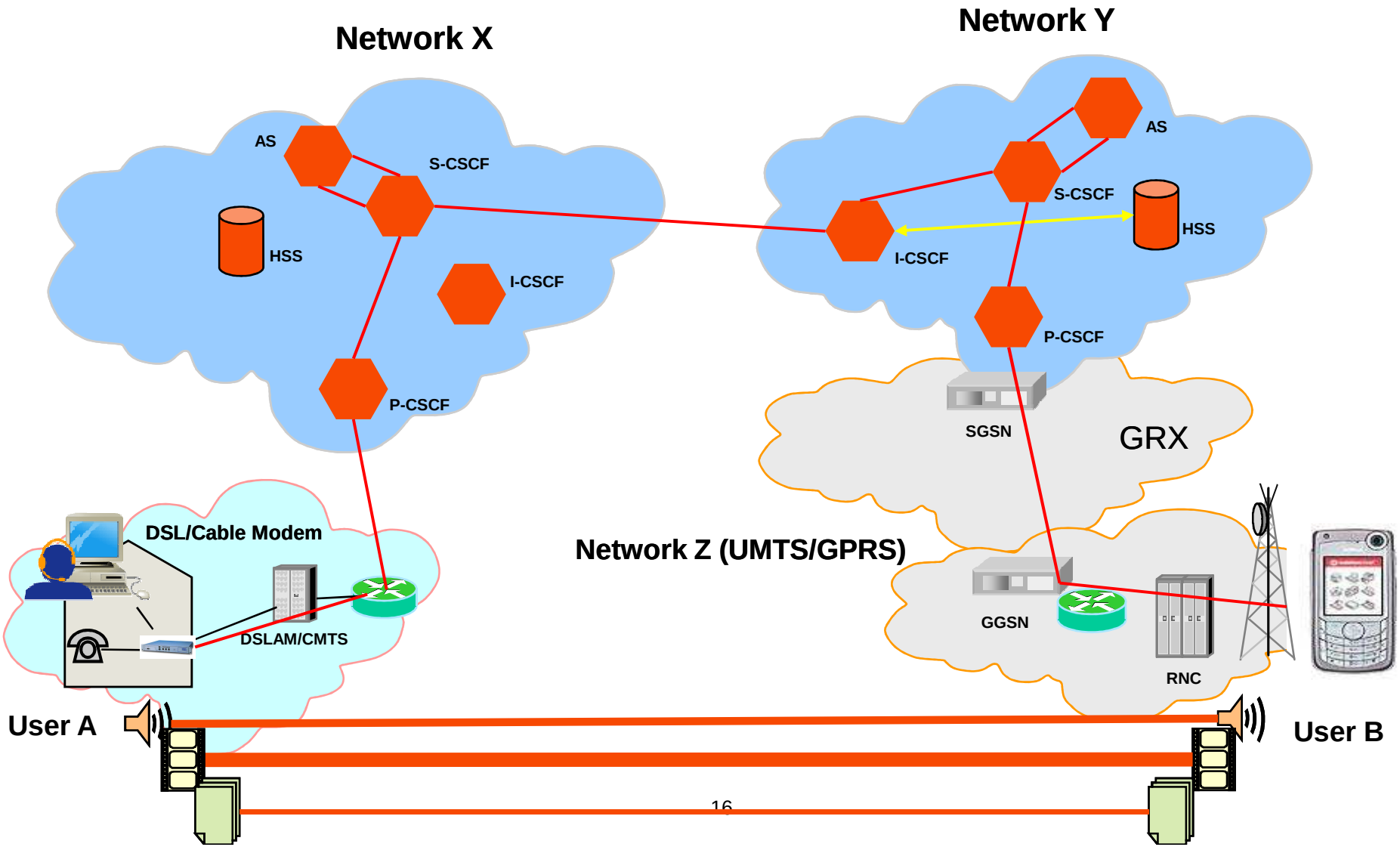
IMS Subscription to UE State Changes



- The P-CSCF Subscribes to the UA/UE Registration State
- S-CSCF Notifies the P-CSCF of Registration State
- The UA/UE Subscribes to its Registration State
- S-CSCF Notifies the UA/UE of Registration State

Now the Elements can Inform Each Other of Registration State Changes

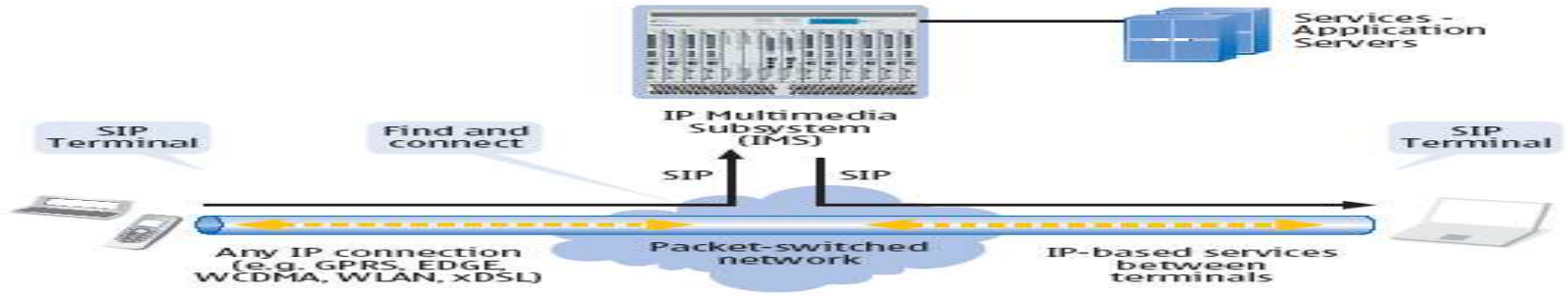
A Typical Example of an IMS Call



Session Initiation Protocol (SIP) in IMS

- Additional Signaling Information
  - For example Cell-ID, Mobile Network/Country Code, Charging-IDs
  - Information transported P-header based solution
- Compression
  - SIP Compression is mandatory as radio interface is a scarce resource
  - Compression / decompression of SIP will be performed by the UE and the P-CSCF
- Authentication & Integrity protection
  - S-CSCF performs the Authentication using AKA
  - P-CSCF checks the integrity of messages received via the air interface via IPsec ESP

SIP based session management



SIP Message Types

Requests – Sent from client to server

- INVITE
- ACK
- REFER
- OPTIONS
- BYE
- CANCEL
- REGISTER
- SUBSCRIBE
- NOTIFY
- MESSAGE

SIP Message Types (Contd.)

Responses – Sent from server to the client

- Success
- Redirection
- Forwarding
- Request failure
- Server failure
- Global failure

Experiences and Challenges

- Charging
- IPv4/v6 Interworking
- Identity Management and USIM/ISIM Migration
- Security and Authentication

Charging

- Two fundamental charging approaches.
  - Offline charging (e.g. pay for bill)
  - Online charging (e.g. pre-pay credit)
- Other charging mechanisms:
  - Session-based charging
  - Event-based charging
  - Flow-based charging

IMS – Security Challenges

IMS - Mobile 2 Mobile Security

- 3GPP did not account for it in the design,
- GSMA identified the problem:
- IMS introduces Mobile to Mobile traffic.
- GPRS was not intended for that
- The problem : difficult to control M2M traffic

3GPP Release 5 Security

- Packet Switched (PS) domain
  - access security features retained from 3GPP Release 99 specifications
- IP Multimedia Subsystem (IMS) domain
  - new **access security** features to be specified
    - to protect the access link to the IMS domain
    - independent of underlying PS domain security features
  - **network domain security** features to protect signalling links between network elements with the IMS domain

Access Security: Authentication Principles

- 3GPP authentication protocol (3GPP AKA)
  - based on secret key stored in UA's tamper-proof subscriber identity module (SIM) and in the HSS
- Authentication check located in S-CSCF
- Working assumption is to authenticate only at SIP registrations with on-demand re-authentication requiring re-registration
- Use SIP authentication rather than an outer layer protocol such as TLS or IKE in order to minimise roundtrips

Integration of Authentication Protocol into DIAMETER and SIP

- Distribution of authentication information to S-CSCF using DIAMETER
  - distribution of authentication vectors for 3GPP AKA
- Integration of authentication protocol into SIP registration
  - 3GPP AKA protocol between UA and S-CSCF
  - distribution of session key to P-CSCF

Access Security: Security Mode Establishment between UA and P-CSCF

- Determines when to start applying protection and which algorithm to use
  - includes secure algorithm negotiation
- Uses session key derived during authentication
- Integration into SIP registration with no new roundtrips

Access security: Protection of SIP signalling between UA and P-CSCF

- Integrity protection of SIP signalling between UA and P-CSCF
- Uses session key derived during authentication
- Symmetric scheme because of efficiency concerns
- Candidate mechanisms include modified CMS and ESP

Other IP Multimedia Subsystem Security Issues (1)

- Hide caller's public ID from called party
  - by encrypting remote party ID header at caller's S-CSCF and decrypting by same S-CSCF
  - is there a requirement to hide caller's IP addresses that are dynamically assigned?
- Network configuration hiding
  - mechanism being developed to hide host domain name of CSCFs and number of CSCFs within one operator's network

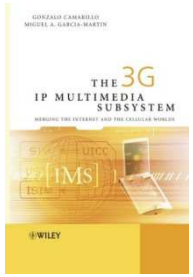
Other IP Multimedia Subsystem Security Issues (2)

- Session transfer
  - guidance on security aspects based on GSM call transfer feature
    - authorisation and accounting of transferred leg needs to involve transferring party who has dropped out of session
    - should there be a limit to the number of transferred sessions?
    - should final destination be hidden from calling party?
- Security aspects of other IP multimedia subsystem services?
- End-to-end security

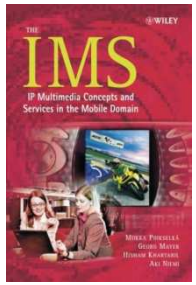
Typical SIP attacks

- Malformed Message Attacks
- Buffer Overflow Attacks
- Denial-of Service attacks
- RTP session hijacking
- Injection of unauthentic RTP packets into existing RTP flows
- Re-use of compromised SIP credentials
- Hostile SIP network elements

Books



- G. Camarillo, M. A. Garcia-Martin: “The 3G IP Multimedia Subsystem (IMS): Merging the Internet and the Cellular Worlds". Jon Willey & Sons, 2004.



- M. Poikselka, G. Mayer, H. Khartabil, A. Niemi : “The IMS – IP Multimedia Concepts and Services in the Mobile Domain”, Jon Willey & Sons, 2004